Preparing High Fidelity Thermofield Double States

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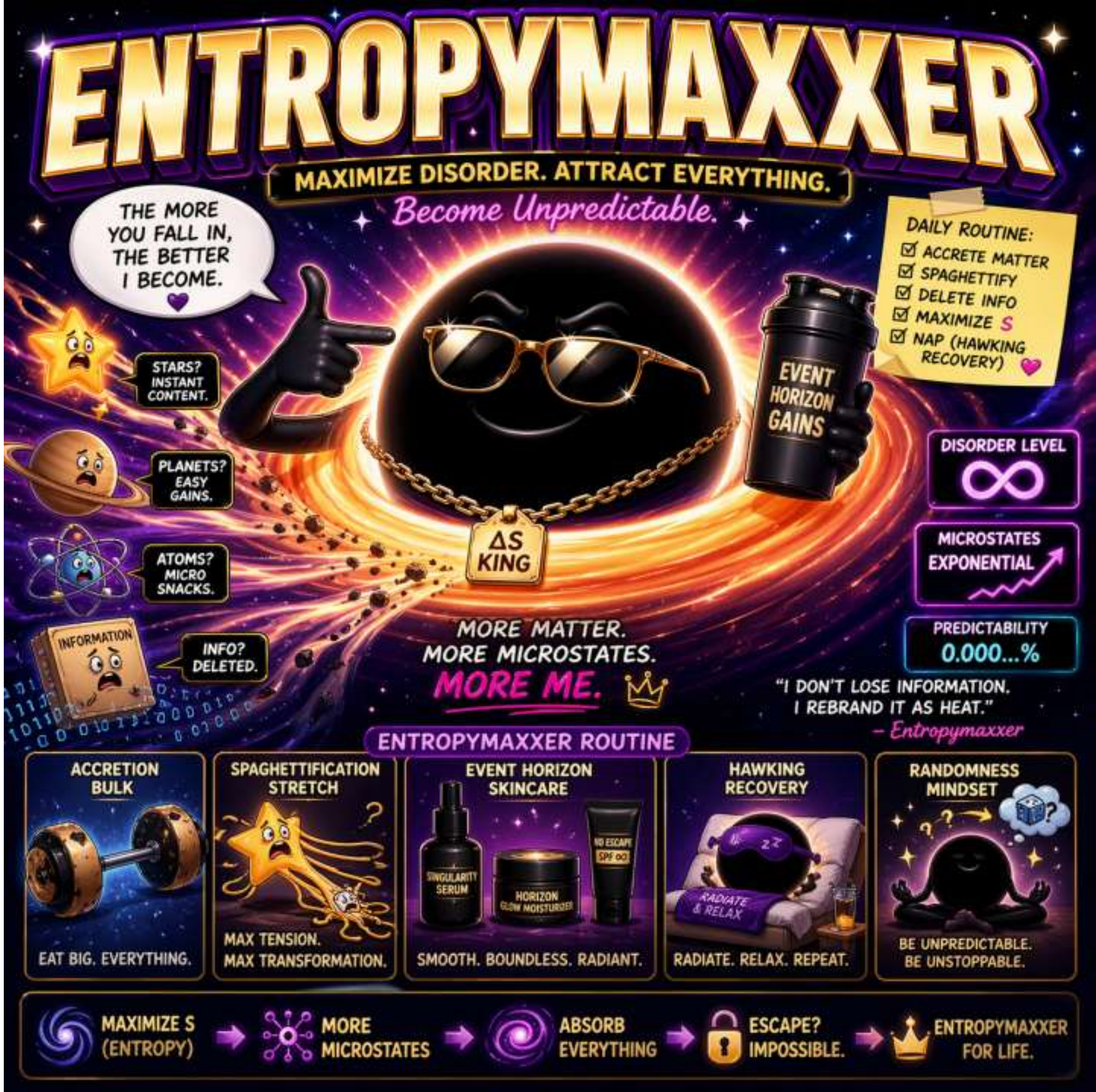

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Preparing High Fidelity Thermofield Double States

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Thermal equilibrium

- Many physical systems can be idealized as existing in thermal equilibrium; local thermal equilibrium is even more common
- If we could prepare thermal states efficiently on a quantum computer, we could do a lot of interesting work
- For example:
  - We could study basic science problems, e.g. high temp superconductors, spin liquids, quark gluon plasma, holographic black holes …
  - We could search for novel materials with desirable electronic, magnetic, etc. properties
  - We might open the door to novel optimization algorithms, e.g. by recasting optimization problem as an energy minimization

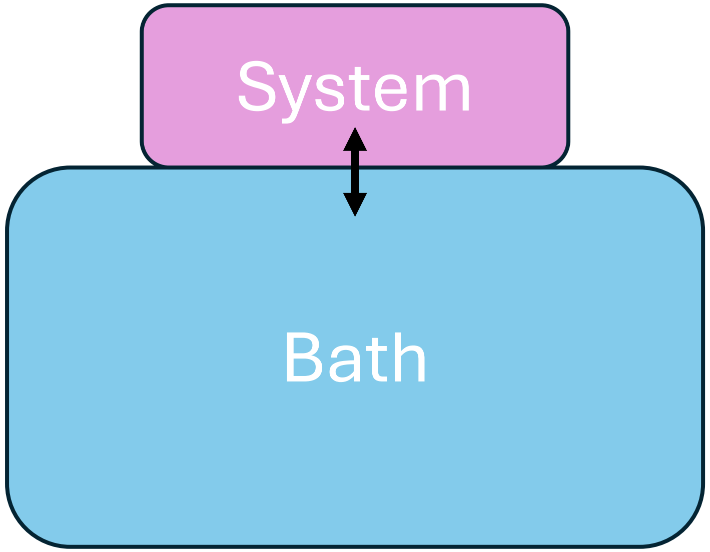
Preparing thermal states

- Given the wide-ranging nature of these problems, we naturally want to have general purpose algorithms to efficiently prepare thermal states

$$H, \beta = \hbar / k_B T \longrightarrow \text{Quantum Algorithm} \longrightarrow \rho = \frac{e^{-\beta H}}{Z}$$

Is this even possible?

- Nature provides us with one relatively general algorithm: couple the system of interest to a heat bath at the desired temperature and let the system come into equilibrium with the bath
- This is general purpose but it may take a long time to equilibrate*



$$\rho_S(t \to \infty) \approx e^{-\beta H}/Z$$

*This is a real limitation, for example:

- Glasses may not equilibrate on any reasonable timescale
- Critical slowing down at a continuous phase transition
- Encoding of hard optimization problems suggests a barrier

Thermofield double states

- Sometimes we want something stronger: a special structured purification of the thermal state, a thermofield double state
- This state shows up in quantum gravity, field theory, and many-body physics in various important ways
- Nature's algorithm does NOT provide this

$H, \beta \longrightarrow$ Quantum Algorithm $\longrightarrow |\mathrm{TFD}\rangle = \sum_n \sqrt{\frac{e^{-\beta E_n}}{Z}} |E_n\rangle_L |E_n\rangle_R$

What is known?

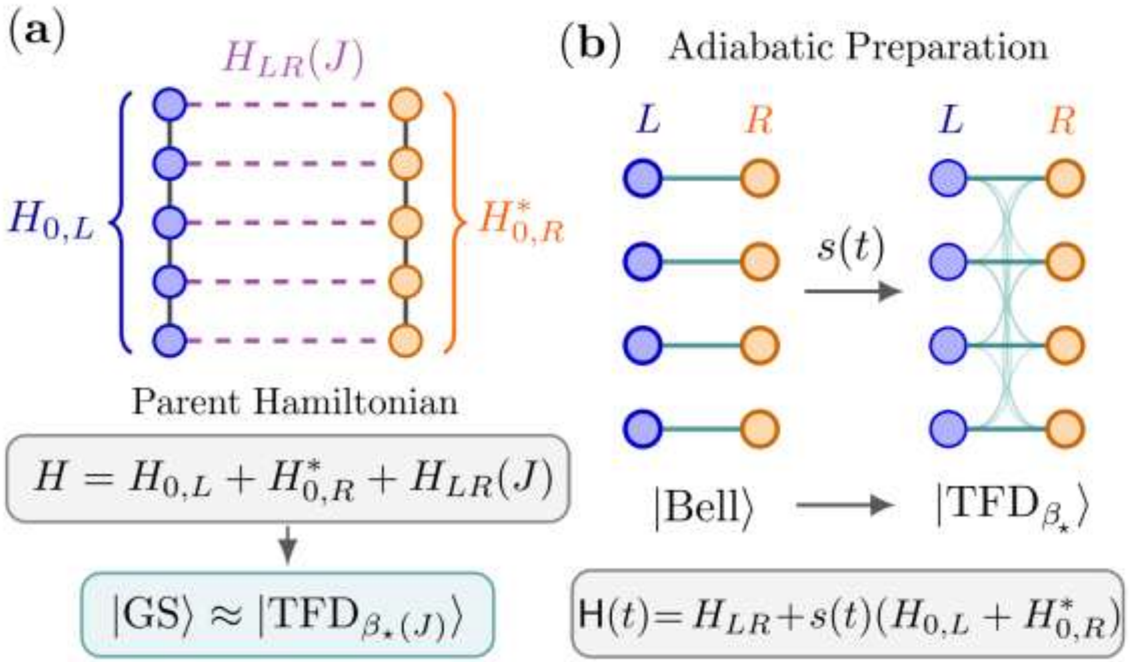
For mixed thermal states, there are a variety of approaches:

- Open system dynamics
- Sampling methods (e.g. Chen et al. ‘23, $O(\beta)$ footprint Lindblads)
- Variational methods
- Adiabatic methods
- “Imaginary” time methods
- (whole host of classical methods as well)

Some of these can be adapted to address the TFD

Our protocol

$$H = H_{0,L} + H^*_{0,R} + \frac{JN}{2K}\sum_{\alpha=1}^{K} \left(\mathcal{O}_{\alpha,L} - \mathcal{O}^*_{\alpha,R}\right)^2 ,$$



- Part 1: Construct a Hamiltonian family whose GS is an approximate TFD at a tunable temp (controlled by LR coupling), adiabatic preparation
- Part 2: Improved fidelity by adding a penalty term that favors a diagonal subspace, $|E_n\rangle_L |E_n\rangle_R$

We provide evidence that this will work when the system self-thermalizes

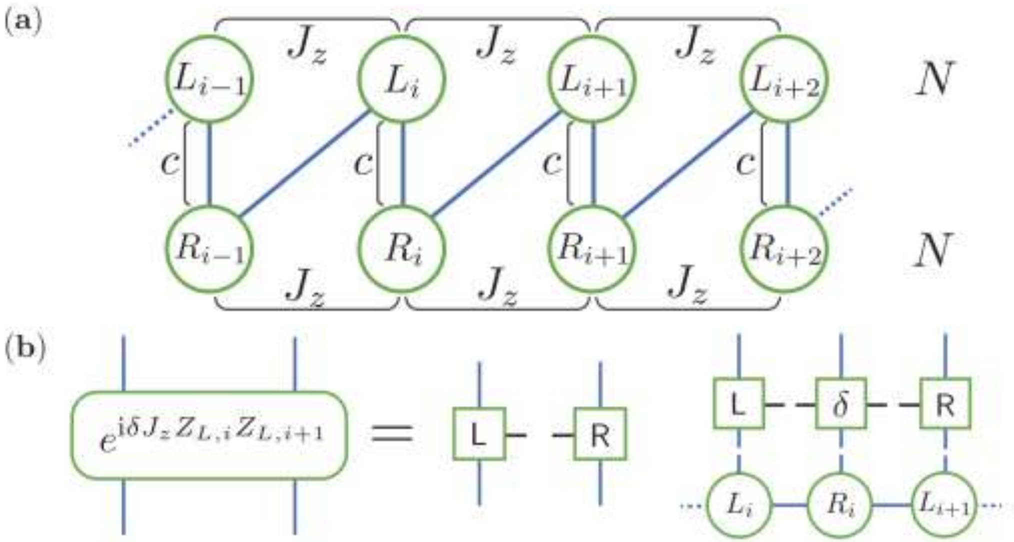
Background for our approach

- There are a few early works e.g. Feiguin-Klich ‘13 and McGreevy-S ‘16 that looked for parent Hamiltonians for TFD states
- A breakthrough came with Maldacena-Qi ’18 (MQ) who proposed a simple approach for SYK and studied its gravity dual
- Cottrell et al. ‘18 suggested that MQ generalized to systems obeying eigenstate thermalization, but some issues emerged
- Bentsen-Nguyen-S ‘23 and Schuster-Kobrin-Su-Marrochio-Yao ‘25 discussed an adiabatic approach to state preparation in SYK
- In our paper, we show concretely how to generalize MQ to spins etc., clarify the issues in Cottrell et al., and provide solutions

Example: mixed field Ising chain

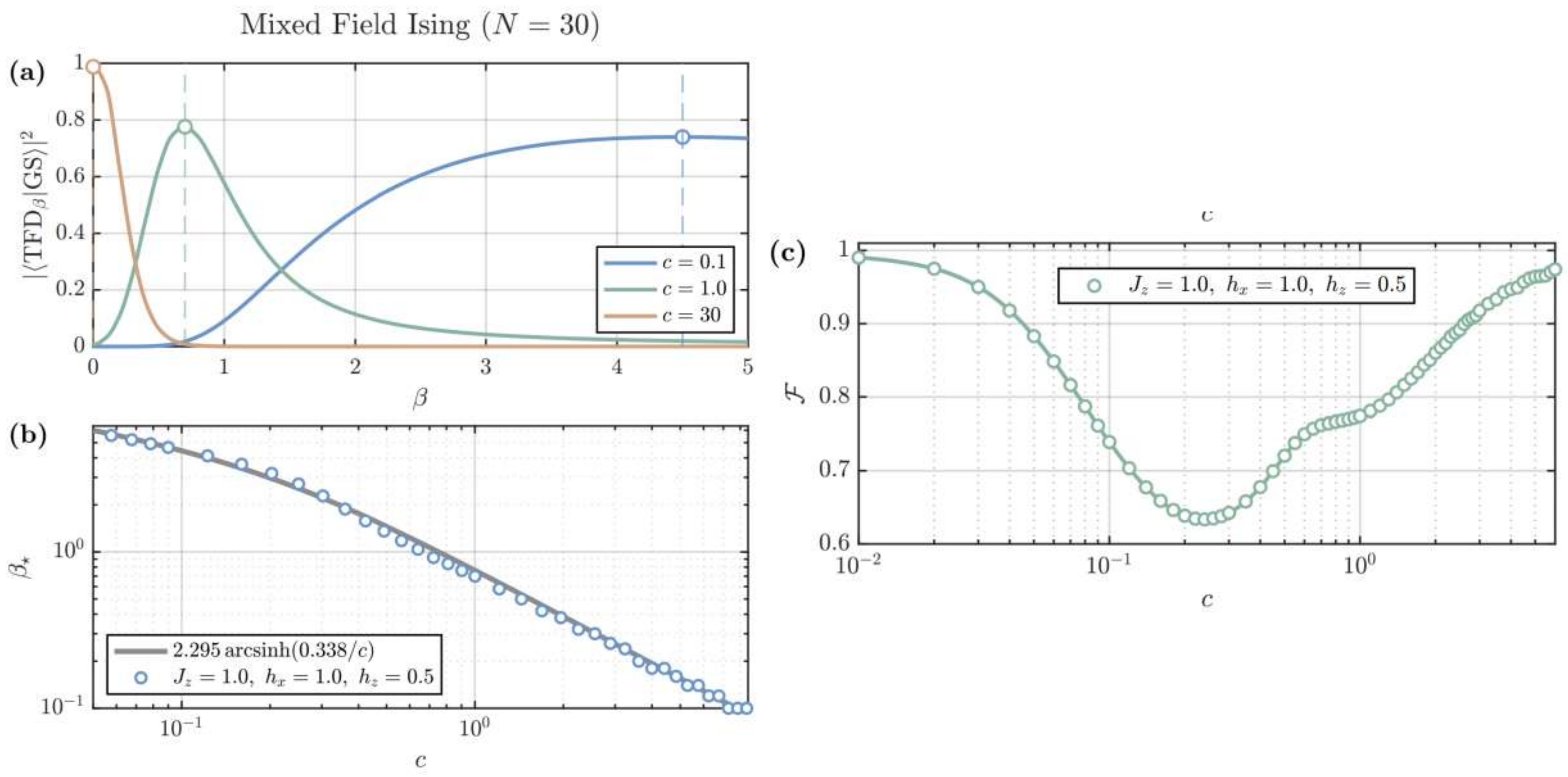
$$H_{\mathrm{MFI},L/R} = J_z \sum_{i=1}^{N-1} Z_{i,L/R} Z_{i+1,L/R} + h_x \sum_{i=1}^{N} X_{i,L/R} + h_z \sum_{i=1}^{N} Z_{i,L/R}\,,$$

$$H = H_{\mathrm{MFI},L} + H^*_{\mathrm{MFI},R} + c \sum_{i=1}^{N} (X_{i,L} - X_{i,R})^2 + c \sum_{i=1}^{N} (Z_{i,L} - Z_{i,R})^2\,,$$

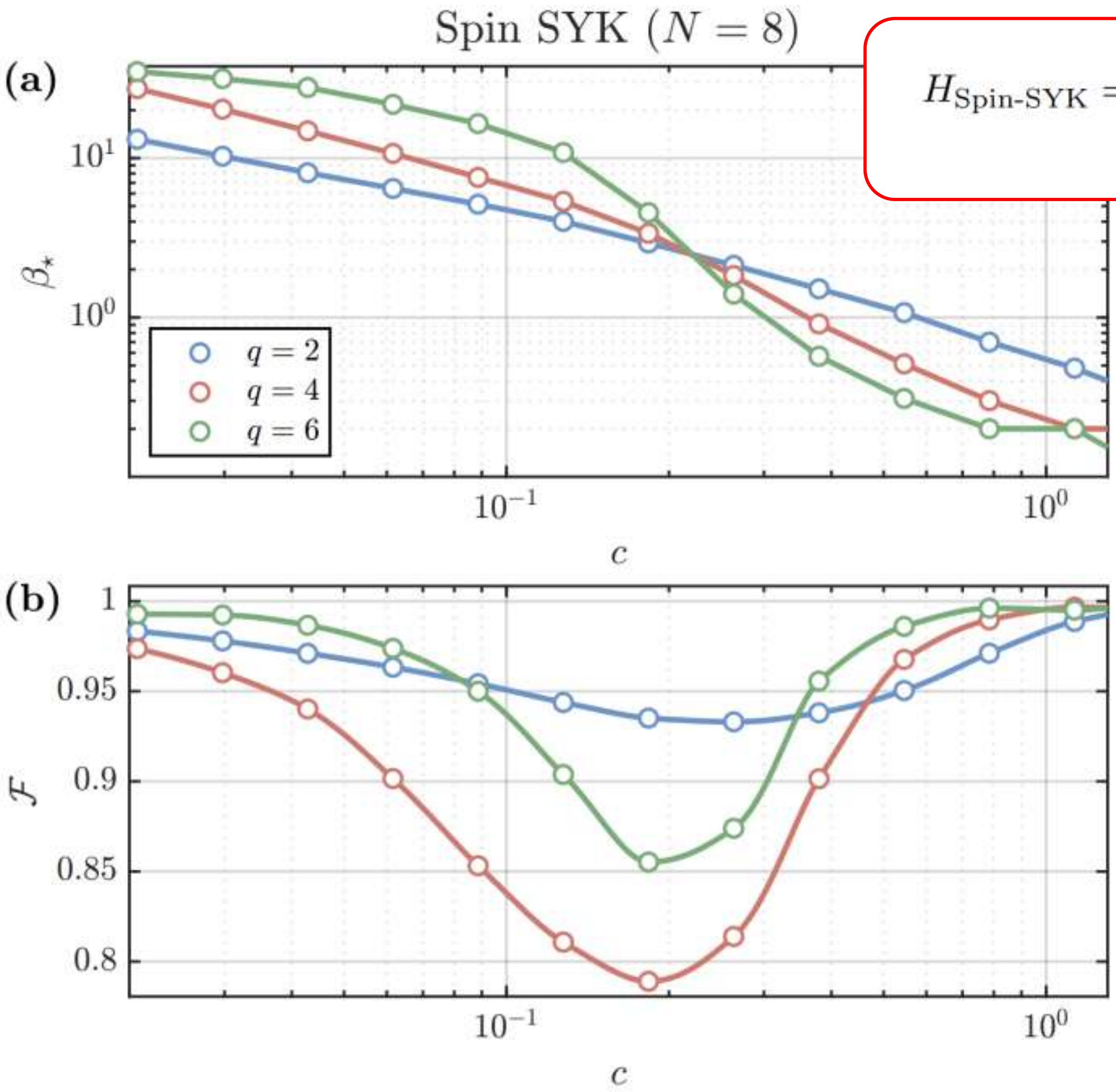


c = LR coupling, need both X and Z couplings

We use MPS tech to compute

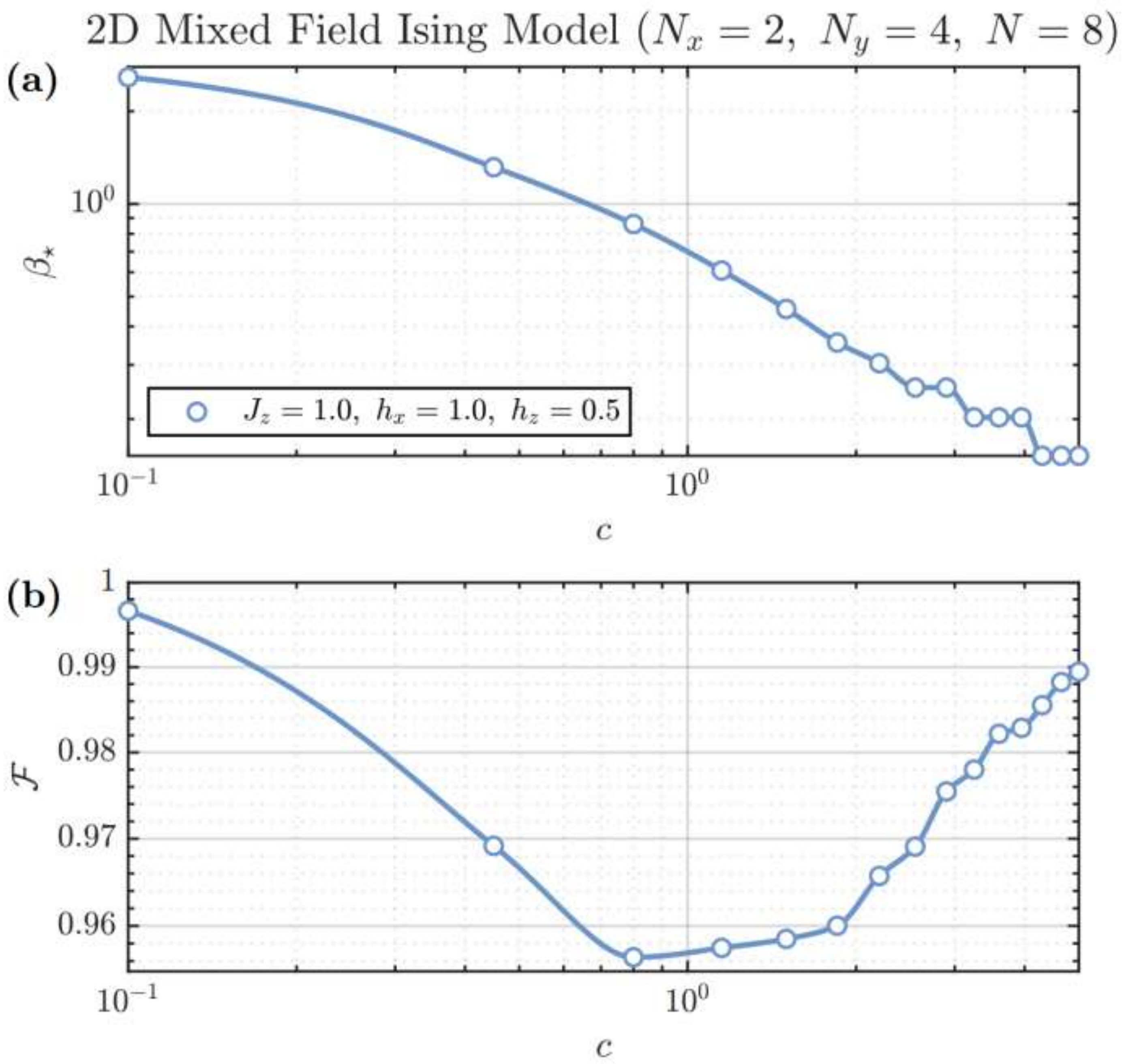


Mixed Field Ising ($N = 30$)
(a)
$|\langle \mathrm{TFD}_\beta | \mathrm{GS} \rangle|^2$
β
$c = 0.1$
$c = 1.0$
$c = 30$
(b)
$\beta_\star$
c
$2.295\,\mathrm{arcsinh}(0.338/c)$
$J_z = 1.0,\ h_x = 1.0,\ h_z = 0.5$
(c)
c
$\mathcal{F}$
$J_z = 1.0,\ h_x = 1.0,\ h_z = 0.5$
c



$$H_{\text{Spin-SYK}} = \sum_{i_1<\cdots<i_q} \sum_{\substack{\alpha_1,\ldots,\alpha_q\in\{x,y\} \\ \#(\alpha=y)\ \text{even}}} J_{i_1\cdots i_q}^{\alpha_1\cdots\alpha_q} \sigma_{i_1}^{\alpha_1}\sigma_{i_2}^{\alpha_2}\cdots\sigma_{i_q}^{\alpha_q},$$

[S-Winer, Tezuka et al.]

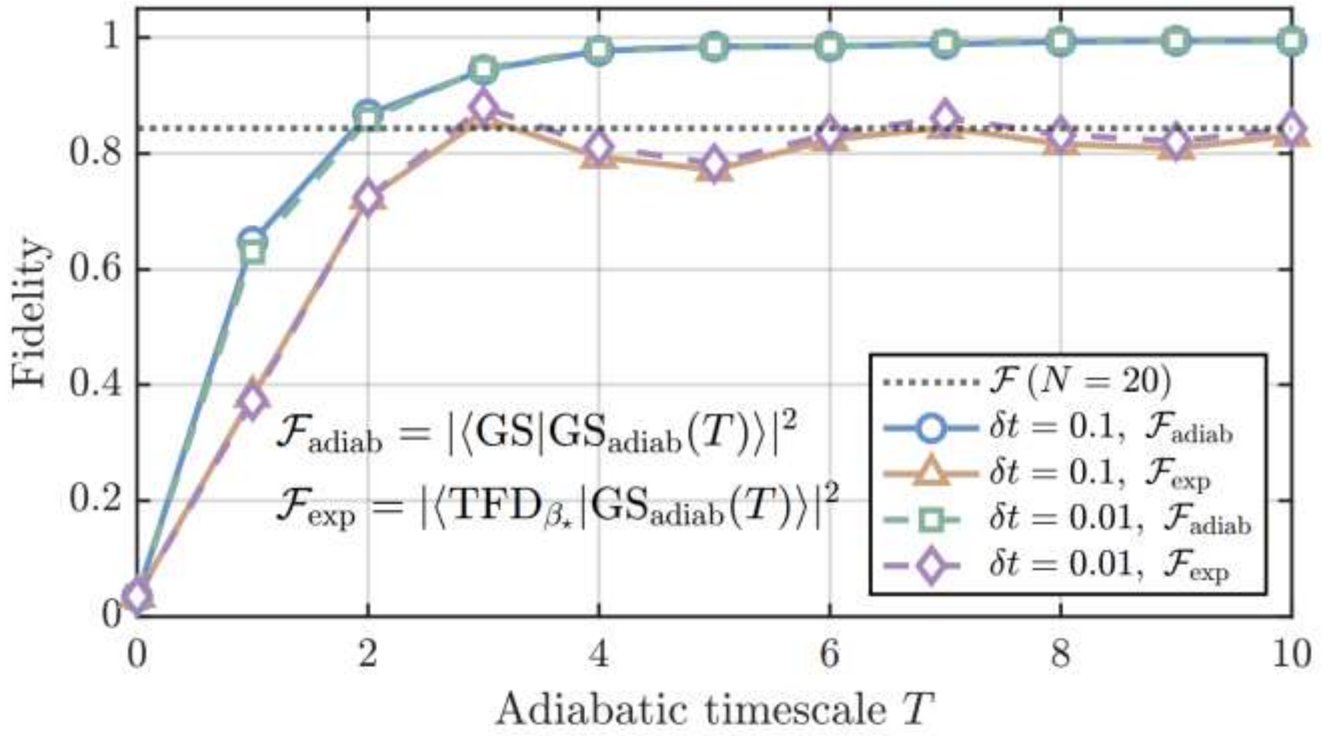


2D Mixed Field Ising Model ($N_x = 2,\ N_y = 4,\ N = 8$)
(a)
10^0
$\beta_\star$
$J_z = 1.0,\ h_x = 1.0,\ h_z = 0.5$
10^{-1}
10^0
c
(b)
1
0.99
0.98
0.97
0.96
$\mathcal{F}$
10^{-1}
10^0
c

Adiabatic protocol

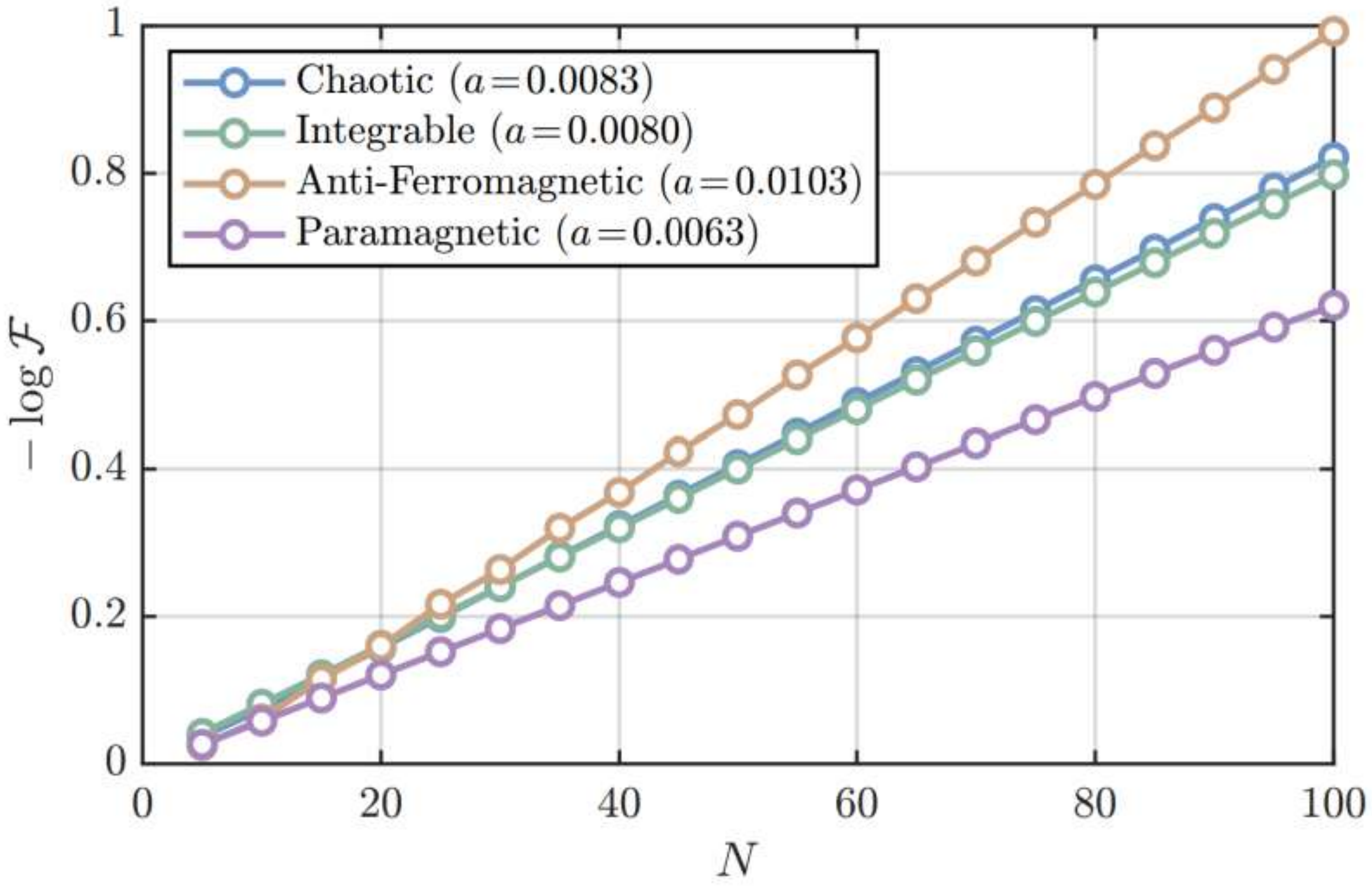
$$\mathsf{H}(t) = H_{LR} + s(t)\big(H_{\mathrm{MFI},L} + H^{*}_{\mathrm{MFI},R}\big)\,,$$

$$s(m) = \left(m - \frac{1}{2}\right)\frac{\delta t}{T}, \qquad m = 1, \ldots, n\,,$$



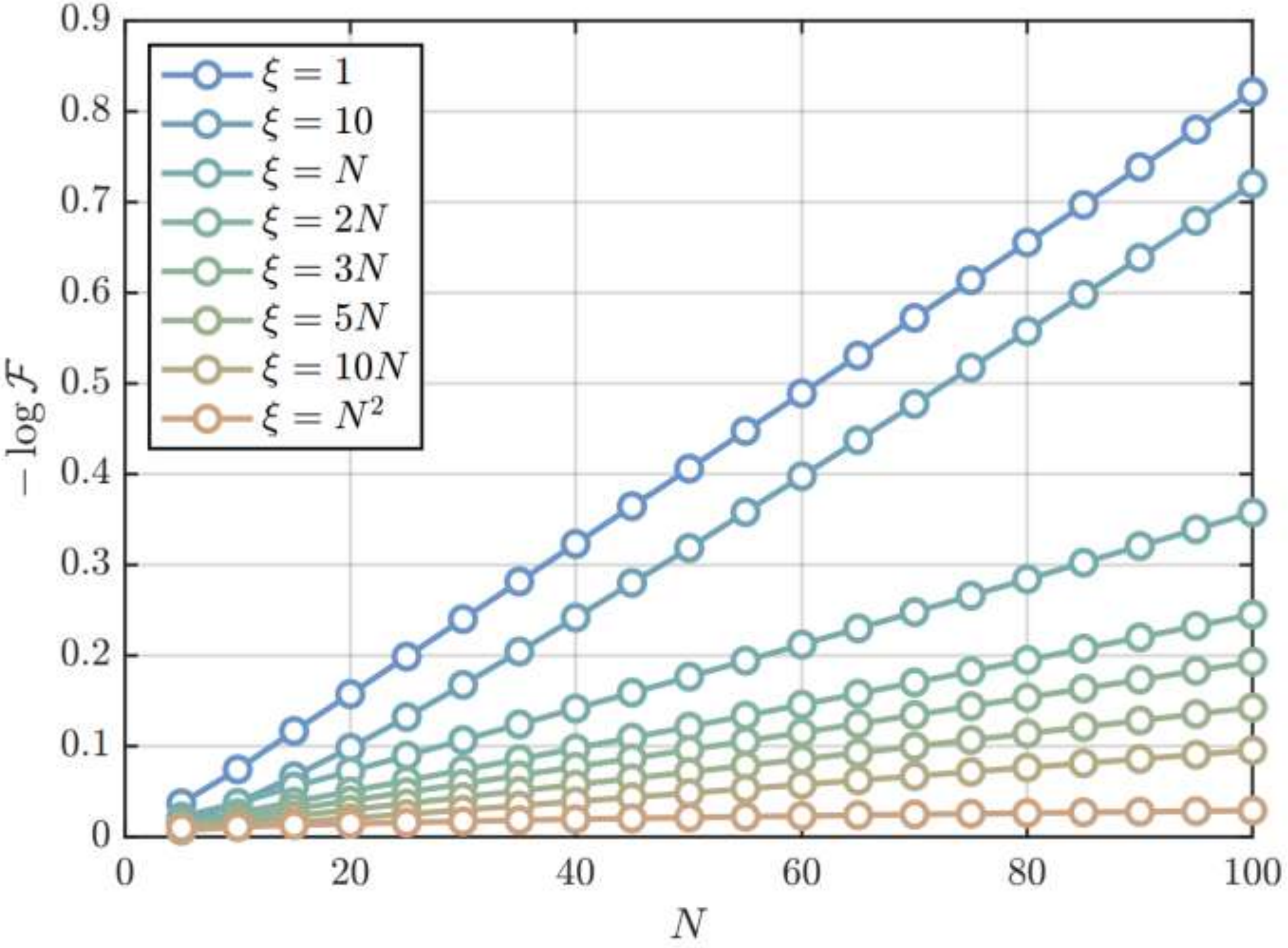
$$\begin{aligned} U(s) \;=\; & U_{1\text{-}body}\left(s, h_x, h_z, \frac{\delta t}{2}\right) U_{ZZ}\left(s, J_z, \frac{\delta t}{2}\right) \\ & \times U_{LR}(c, \delta t)\, U_{ZZ}\left(s, J_z, \frac{\delta t}{2}\right) \\ & \times U_{1\text{-}body}\left(s, h_x, h_z, \frac{\delta t}{2}\right)\,, \end{aligned}$$

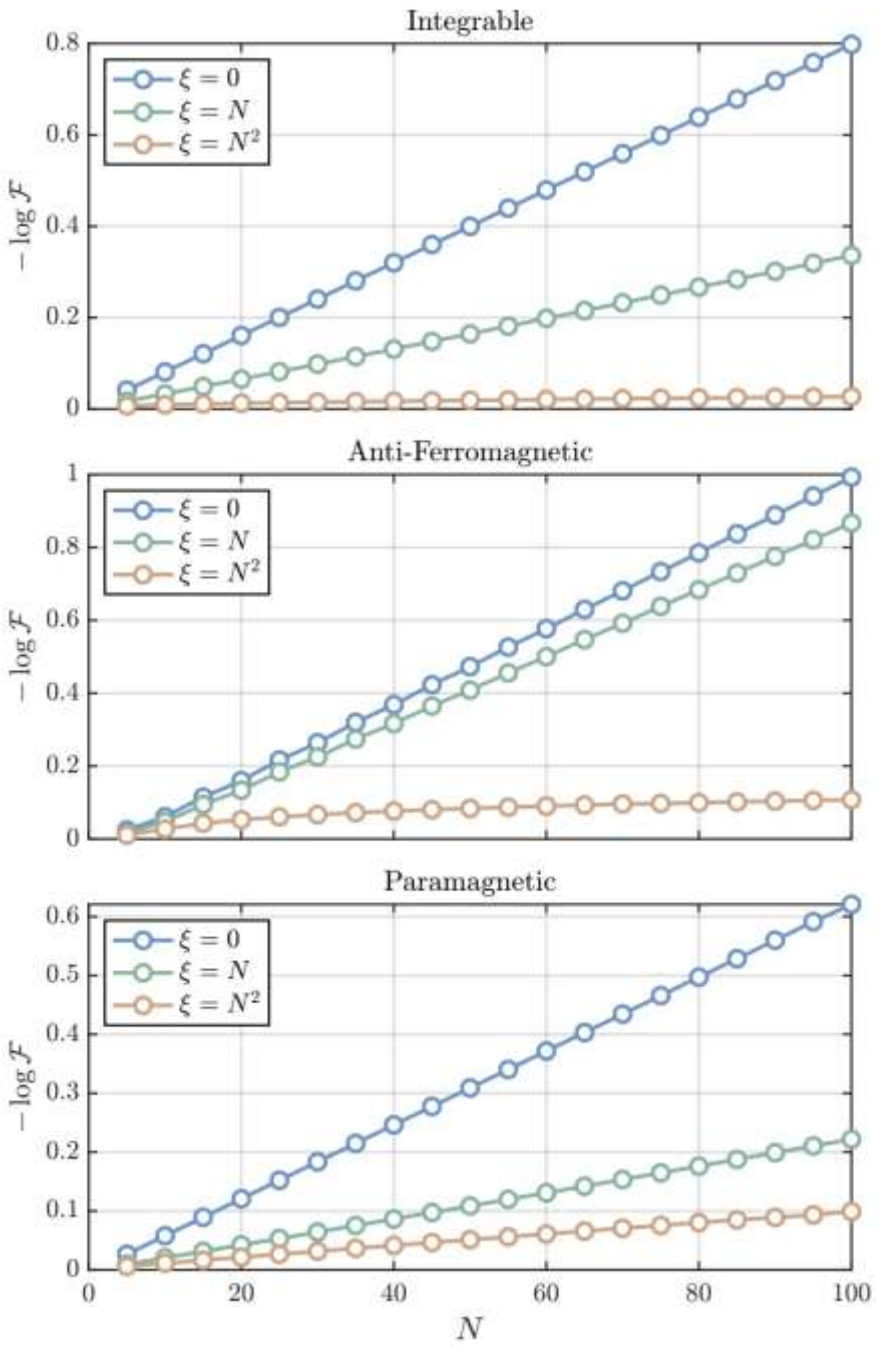
Scaling the system size



Penalty term

$$\frac{\xi}{2N}(H_{0,L} - H^*_{0,R})^2,$$





Taking $\xi \sim N^a$ is unusual scaling for a term in a natural Hamiltonian, but it presents no obstacle to a poly(N) quantum algorithm

Gravity calculation + general argument

- For the SYK model in its low-energy regime, we can do some further analytics to get (e.g. $\Delta_{\mathcal{O}} = 1/4$):

$$\log \mathcal{F}\,|_{\xi} \sim -N \times (\text{small coefficient}) \times (N/\xi)^{2-4\Delta_{\mathcal{O}}},$$

$$\mathcal{J}\xi \gtrsim N^{\frac{3-4\Delta_{\mathcal{O}}}{2-4\Delta_{\mathcal{O}}}}.$$

- A general perturbative calculation suggests $\xi \gtrsim N^3$ may be sufficient to stick in the diagonal subspace

$$\|P_{\text{off}}|\text{GS}\rangle\|^2 \sim \frac{N^2}{K}\int d\omega\, \frac{\hat{\rho}_{\text{off}}(\omega)}{\left(\Delta_{\text{off}} + \xi\omega^2/N\right)^2},$$

Outlook

- We presented a candidate general purpose algorithm for TFD prep and gave evidence that it works for a variety of Hamiltonians
- For near term applications, an adiabatic approach prepares a high fidelity TFD state at modest cost
- For devices with logical qubits, one can add a penalty term to prepare large scale high fidelity TFD states
- Can we understand more deeply when and how the algorithm works?

Learning? (preliminary)

- The GS of the LR Hamiltonian is not exactly a TFD of the original Hamiltonian, so what is it?
- Idea: it is the TFD of a "nearby" Hamiltonian
  - $H_{MQ}[H_0] \to |\mathrm{TFD}[H_0 + \Delta H]\rangle$
  - Adjust H_0 such that $H_0 + \Delta H = H_{\mathrm{target}}$?

Learn a better H_0 by maximizing similarity with local state of a small exact thermal state; we allow H_0 to have all 2-body terms:

$$(\mathrm{XX}, \mathrm{XY}, \mathrm{XZ}, \mathrm{YX}, \mathrm{YY}, \mathrm{YZ}, \mathrm{ZX}, \mathrm{ZY}, \mathrm{ZZ}, \mathrm{X}, \mathrm{Y}, \mathrm{Z})$$